

Green's Theorem

Green's theorem relates a line integral to the double integral taken over the region bounded by the closed curve.

Statement

If $M(x, y)$ and $N(x, y)$ are continuous functions with continuous, partial derivatives in a region R of the xy - plane bounded by a simple closed curve C , then

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy, \text{ where } C \text{ is the curve described in the positive direction.}$$

Vector form of Green's theorem

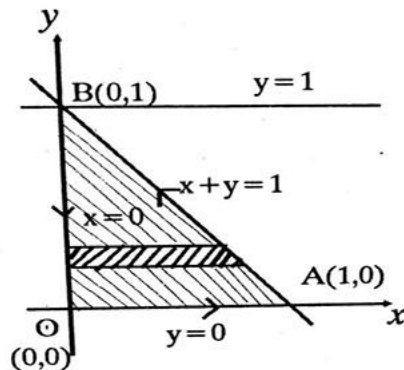
$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{k} dR$$

Problems based on Green's theorem

Example 1.

Verify Green's theorem in the plane for $\int (3x^2 - 8y^2)dx + (4y - 6xy)dy$, along C , where C is the boundary of the region $x = 0, y = 0, x + y = 1$.

Solution:



$$\text{We have to prove that } \int_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Here, $M = 3x^2 - 8y^2$ and $N = 4y - 6xy$

$$\Rightarrow \frac{\partial M}{\partial y} = -16y \quad \Rightarrow \frac{\partial N}{\partial x} = -6y$$

$$\therefore \int_C (3x^2 - 8y^2)dx + (4y - 6xy)dy = \int_C M dx + N dy$$

By Green's theorem in the plane,

$$\begin{aligned}
 \int_c M dx + N dy &= \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy \\
 &= \int_0^1 \int_0^{1-x} (10y) dy dx \\
 &= 10 \int_0^1 \left[\frac{y^2}{2} \right]_0^{1-x} dx \\
 &= 5 \int_0^1 (1-x)^2 dx \\
 &= 5 \left[\frac{(1-x)^3}{-3} \right]_0^1 = \frac{5}{3} \dots (1)
 \end{aligned}$$

Consider $\int M dx + N dy = \int_{OA} + \int_{AB} + \int_{BO}$

Along $OA, y = 0 \Rightarrow dy = 0, x$ varies from 0 to 1

$$\therefore \int_{OA} M dx + N dy = \int_0^1 3x^2 dx = [x^3]_0^1 = 1$$

Along $AB, y = 1 - x \Rightarrow dy = -dx$ and x varies from 1 to 0

$$\begin{aligned}
 \therefore \int_{AB} M dx + N dy &= \int_1^0 [3x^2 - 8(1-x)^2 - 4(1-x) + 6x(1-x)] dx \\
 &= \left[\frac{3x^3}{3} - \frac{8(1-x)^3}{-3} - \frac{4(1-x)^2}{-2} + 3x^2 - 2x^3 \right]_1^0 \\
 &= \frac{8}{3} + 2 - 1 - 3 + 2 = \frac{8}{3}
 \end{aligned}$$

Along $BO, x = 0 \Rightarrow dx = 0$ and y varies from 1 to 0

$$\therefore \int_{BO} M dx + N dy = \int_1^0 4y dy = [2y^2]_1^0 = -2$$

$$\therefore \int_c M dx + N dy = 1 + \frac{8}{3} - 2 = \frac{5}{3} \dots (2)$$

$\therefore$ From (1) and (2)

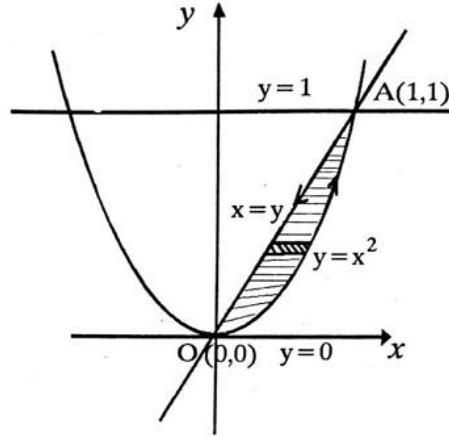
$$\therefore \int_c M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Hence Green's theorem is verified.

Example 2.

Verify Green's theorem in the XY plane for $\int (xy + y^2)dx + x^2dy$, along C, where C is the closed curve of the region bounded by $x = y, y = x^2$.

Solution:



We have to prove that $\int_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$

Here, $M = xy + y^2$ and $N = x^2$

$$\Rightarrow \frac{\partial M}{\partial y} = x + 2y \quad \Rightarrow \frac{\partial N}{\partial x} = 2x$$

$$\text{R.H.S} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Limits:

x varies from y to $\sqrt{y}$

y varies from 0 to 1

$$\begin{aligned} \therefore \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy &= \int_0^1 \int_y^{\sqrt{y}} 2x - (x + 2y) dx dy \\ &= \int_0^1 \left[\frac{x^2}{2} - 2xy \right]_y^{\sqrt{y}} dy \\ &= \int_0^1 \left(\frac{y}{2} - 2y\sqrt{y} \right) - \left(\frac{y^2}{2} - 2y^2 \right) dy \\ &= \int_0^1 \left(\frac{y}{2} - 2y^{\frac{3}{2}} + 3\frac{y^2}{2} \right) dy \\ &= \left[\frac{y^2}{2} - \frac{4y^{\frac{5}{2}}}{5} + \frac{y^3}{2} \right]_0^1 \\ &= \frac{1}{4} - \frac{4}{5} + \frac{1}{2} = -\frac{1}{20} \end{aligned}$$

$$\text{L.H.S} = \int_c M dx + N dy$$

$$\text{Consider } \int M dx + N dy = \int_{OA} + \int_{AO}$$

Along $OA, y = x^2 \Rightarrow dy = 2x dx, x$ varies from 0 to 1

$$\begin{aligned} \therefore \int_{OA} M dx + N dy &= \int_0^1 [(x(x^2) + (x^2)^2)dx + x^2 \cdot 2x dx] \\ &= \int_0^1 (3x^3 + x^4) dx \\ &= \left[\frac{3x^4}{4} + \frac{x^5}{5} \right]_0^1 \\ &= \frac{3}{4} + \frac{1}{5} = \frac{19}{20} \end{aligned}$$

Along $AO, y = x \Rightarrow dy = dx$ and x varies from 1 to 0

$$\begin{aligned} \therefore \int_{AO} M dx + N dy &= \int_1^0 (x^2 + x^2)dx + x^2 dx \\ &= \int_1^0 3x^2 dx = [x^3]_1^0 = -1 \end{aligned}$$

$$\text{L.H.S} = \int_c M dx + N dy = \frac{19}{20} - 1 = -\frac{1}{20}$$

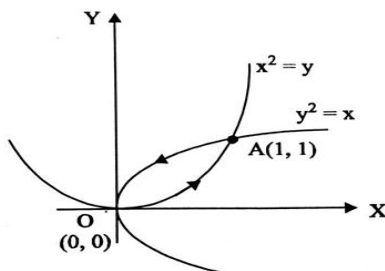
$$\therefore \text{L.H.S} = \text{R.H.S}$$

Hence Green's theorem is verified.

Example 3.

Verify Green's theorem in the plane for $\int (3x^2 - 8y^2)dx + (4y - 6xy)dy$, along C , where C is the boundary of the region defined by $y = x^2, x = y^2$.

Solution:



We have to prove that $\int_c M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$

Here, $M = 3x^2 - 8y^2$ and $N = 4y - 6xy$

$$\Rightarrow \frac{\partial M}{\partial y} = -16y \quad \Rightarrow \frac{\partial N}{\partial x} = -6y$$

$$\text{R.H.S} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Limits:

x varies from y^2 to $\sqrt{y}$

y varies from 0 to 1

$$\begin{aligned} \therefore \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy &= \int_0^1 \int_{y^2}^{\sqrt{y}} (-6y + 16y) dx dy \\ &= \int_0^1 [10xy]_{y^2}^{\sqrt{y}} dy \\ &= 10 \int_0^1 (y\sqrt{y} - y^3) dy \\ &= 10 \left[\frac{y^{\frac{5}{2}}}{\frac{5}{2}} - \frac{y^4}{4} \right]_0^1 \\ &= 10 \left(\frac{2}{5} - \frac{1}{4} \right) = \frac{3}{2} \end{aligned}$$

$$\text{L.H.S} = \int_c M dx + N dy$$

$$\text{Consider } \int M dx + N dy = \int_{OA} + \int_{AO}$$

Along $OA, y = x^2 \Rightarrow dy = 2x dx, x$ varies from 0 to 1

$$\begin{aligned} \therefore \int_{OA} M dx + N dy &= \int_0^1 (3x^2 - 8x^4) dx + (4x^2 - 6x^3)(2x) dx \\ &= \int_0^1 (3x^2 - 8x^4 + 8x^3 - 12x^4) dx \\ &= \int_0^1 (-20x^4 + 8x^3 + 3x^2) dx \\ &= \left[-20 \frac{x^5}{5} + 8 \frac{x^4}{4} + 3 \frac{x^3}{3} \right]_0^1 \\ &= -4 + 2 + 1 = -1 \end{aligned}$$

Along $AO, x = y^2 \Rightarrow dx = 2ydy$ and y varies from 1 to 0

$$\begin{aligned}
 \therefore \int_{Ao} M dx + N dy &= \int_1^0 (3y^4 - 8y^2)2ydy + (4y - 6yy^2) dy \\
 &= \int_1^0 (6y^5 - 16y^3 + 4y - 6y^3) dx \\
 &= \int_1^0 (6y^5 - 22y^3 + 4y) dx \\
 &= \left[6\frac{y^6}{6} - 22\frac{y^4}{4} + 4\frac{y^2}{2} \right]_1^0 \\
 &= 0 - \left[1 - \frac{11}{2} + 2 \right] \\
 &= -\left(3 - \frac{11}{2} \right) = \frac{5}{2}
 \end{aligned}$$

$$\text{L.H.S} = \int_c M dx + N dy = -1 + \frac{5}{2} = \frac{3}{2}$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

Hence Green's theorem is verified.

Example 4.

Verify Green's theorem in the plane for the integral $\int (x - 2y)dx + xdy$, taken around the circle $x^2 + y^2 = 1$.

Solution:

$$\text{We have to prove that } \int_c M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Here, $M = x - 2y$ and $N = x$

$$\Rightarrow \frac{\partial M}{\partial y} = -2 \quad \Rightarrow \frac{\partial N}{\partial x} = 1$$

$$\text{R.H.S} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$\begin{aligned}
 \therefore \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy &= \iint_R (1 + 2) dx dy \\
 &= 3 \iint_R dx dy \\
 &= 3 (\text{Area of the circle}) \\
 &= 3\pi r^2 \\
 &= 3\pi \quad (\because \text{radius} = 1)
 \end{aligned}$$

$$\text{L.H.S} = \int_c M dx + N dy$$

Given C is $x^2 + y^2 = 1$

The parametric equation of circle is

$$x = \cos \theta, \quad y = \sin \theta$$

$$dx = -\sin \theta d\theta, \quad dy = \cos \theta d\theta$$

Where θ varies from 0 to 2π

$$\begin{aligned} \therefore \int_c M dx + N dy &= \int_0^{2\pi} (\cos \theta - 2 \sin \theta) (-\sin \theta d\theta) + \cos \theta (\cos \theta d\theta) \\ &= \int_0^{2\pi} (-\sin \theta \cos \theta + 2 \sin^2 \theta + \cos^2 \theta) d\theta \\ &= \int_0^{2\pi} (-\sin \theta \cos \theta + \sin^2 \theta + 1) d\theta \quad (\because \sin^2 \theta + \cos^2 \theta = 1) \\ &= \int_0^{2\pi} \left(-\frac{\sin 2\theta}{2} + \frac{1 - \cos 2\theta}{2} + 1 \right) d\theta \\ &= \left[-\frac{1}{2} \left(-\frac{\cos 2\theta}{2} \right) + \frac{\theta}{2} - \frac{1}{2} \left(\frac{\sin 2\theta}{2} \right) + \theta \right]_0^{2\pi} \\ &= \left[\frac{\cos(4\pi)}{4} + \frac{2\pi}{2} - \frac{\sin 4\pi}{4} + 2\pi \right] - \left[\frac{\cos 0}{4} + \frac{0}{2} - \frac{\sin 0}{4} + 0 \right] \\ &= \frac{1}{4} + \pi + 2\pi - \frac{1}{4} = 3\pi \quad [\because \sin n\pi = 0, \sin 0 = 0, \cos 0 = 1], [\cos n\pi = (-1)^n] \end{aligned}$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

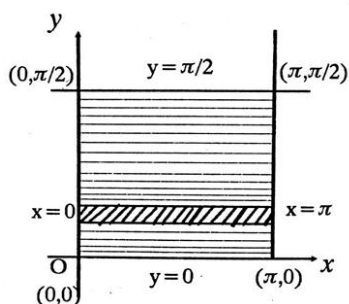
Hence Green's theorem is verified.

Exercise:

Q1. Using Green's theorem, evaluate $\int (y - \sin x) dx + \cos x dy$ where C is the triangle bounded by $y = 0, x = \frac{\pi}{2}, y = \frac{2x}{\pi}$. Ans: $-\left[\frac{\pi}{4} + \frac{2}{\pi}\right]$

Q2. Evaluate by Green's theorem, $\int [e^{-x}(\sin y dx + \cos y dy)]$ where C being the rectangle with vertices $(0,0), (\pi, 0), \left(\pi, \frac{\pi}{2}\right), (0, \pi/2)$.

Hints:



Limits:

x varies from 0 to π

y varies from 0 to $\frac{\pi}{2}$

Ans: $2[e^{-\pi} - 1]$

Q3.

Using Green's theorem, find the value of $\int_c [(xy - x^2)dx + x^2ydy]$ along the closed

curve C formed by $y = 0, x = 1$ and $y = x$

Ans: $-\frac{1}{12}$

Q4.

Verify Green's theorem for $\int_c [(x^2 - y^2)dx + 2xydy]$, where C is the boundary of the

rectangle in the xoy - plane bounded by the lines $x = 0, x = a, y = 0$ and $y = b$.

Ans: Common value = $2ab^2$

Q5.

Verify Green's theorem for $\int_c [(2x - y)dx + (x + y)dy]$, where C is the boundary of the

Circle $x^2 + y^2 = a^2$ in the xoy - plane.

Ans: $2\pi a^2$